

IP-SAKTI

Multilingual, RAG-Based AI Assistant for Intellectual Property
and Regulatory Guidance in Ayurveda

Government of India · Ministry of AYUSH · SIH 2026

Date	7 September 2026
Assessed by	Claude Code AI
Overall Progress	~55–60% Complete
Current Stage	MVP — Citation-Grounded Retrieval

1 · Overall Assessment

Verdict: The core RAG + LLM pipeline works. A functional **MVP demonstrating the concept** is in place. The remaining work is content expansion, UI depth, and multi-agent orchestration — exactly what the staged roadmap targets.

Overall Completion: ~55–60%

Feature-by-Feature Status

Dimension	Status	Assessment
RAG corpus (curated, text-based)	Done	10 files across 7 categories, ~1.2 MB of structured content
ChromaDB vector store	Done	Populated, persistent in my_db/ (~3.8 MB)
Source citation (per-query)	Done	Returns sources[] with category, source, relevance_score
LLM integration (custom endpoint)	Done	localhost:20128/v1 OpenAI-compatible; configured in .env
Voice input (STT)	Done	query-voice endpoint: Sarvam AI → fallback to Google Speech
Voice output (TTS)	Done	Sarvam Bulbul TTS v3 wired into text-to-speech endpoint
Jurisdiction toggle	Partial	Present in corpus metadata; no UI toggle — both answer sets conflated
Formulation classification flow	Partial	Corpus has the decision tree; no interactive wizard
ABS compliance helper	Partial	Content exists in corpus; no dedicated wizard or checklist flow
TKDL / prior-art pointer	Partial	TKDL described in corpus; no pointer to actual database
Multilingual delivery	Partial	Sarvam STT+TTS wired; language codes only map to English
Disclaimer ("information, not legal advice")	Missing	No disclaimer on answers or in UI
Confidence indicator (UI)	Partial	Returns confidence in API; not surfaced in chat widget
Human escalation path	Missing	No escalation mechanism whatsoever
Version-tracked corpus	Missing	No versioning, no changelog, no corpus freshness date
Relational knowledge graph	Not started	No graph DB, no entity linking between statutes/cases/herbs/formulations
Agentic multi-source orchestration	Not started	No multi-step reasoning, no sub-agents
DPDPA / AI security compliance	Not started	No audit log, no data retention policy, no consent flow
Evaluation framework	Not started	No accuracy benchmark, no citation correctness scorer

2 · Development Roadmap — Three Staged Build

The build follows a staged approach: **citation-grounded retrieval MVP** first, then the **knowledge graph and agentic orchestration** layer, then **production hardening**.

Stage 1 — Citation-Grounded Retrieval MVP

Where you are today. Target: 6–8 weeks to complete. **Goal:** Every answer traceable to a source; hallucination minimized.

What's already done

- ✓ Curated corpus — text structured into topics/chunks
- ✓ ChromaDB vector store — populated, persistent
- ✓ Priority search — root data.txt → RAG fallback
- ✓ LLM answer generation — with context injection
- ✓ clean_answer() — strips LLM reasoning artifacts

What needs building

#	Task	Priority	Difficulty
1	Jurisdiction toggle UI (India / International / All switch in chat widget)	High	Medium
2	Split answer display — India panel and International panel visibly separate	High	Medium
3	Per-sentence citation overlay (click sentence → highlights source chunk)	High	High
4	Source panel — show statute / rule / treaty article relied upon	High	Medium
5	Corpus upgrade: 2024 Patents Amendment & 2024 Biodiversity Rules (named but text missing)	High	Medium
6	WIPO GRATK Treaty (2024) content — completely absent from corpus	High	Medium
7	Formulation classifier wizard (Classical / PAM / New Drug / Phytopharm / Aahar / Cosmetic)	High	High
8	ABS compliance checklist wizard (PIC → MAT → NBA approval)	High	High
9	TKDL pointer — link to tkdl.res.in with usage guide	Medium	Low
10	Disclaimer banner — "information, not legal advice" on every answer	High	Low
11	Confidence score badge in chat widget (high / medium / low)	Medium	Low
12	Corpus versioning — data/corpus_version.json (version, date, sources)	Medium	Low
13	Evaluation test set — build from Questions.txt, manual accuracy scoring	Medium	Medium

Stage 2 — Knowledge Graph + Agentic Layer

~8–12 weeks after Stage 1. Uses Neo4j for entity linking; multi-step reasoning agents.

- High** — Neo4j knowledge graph (statutes, rules, sections, cases, herbs, formulations, GIs) — nodes + edges: governs / cites / prevents / conflicts_with
- High** — Multi-step reasoning agent — decompose complex queries into sub-steps (classify → ABS → IP → guidance)
- High** — Sub-agents per domain (Patent, GI, ABS, Drug-regulation)
- High** — Corpus expansion: International treaties (TRIPS Art.27, PCT, Madrid, Hague, Budapest Treaty, EU herbal market access)
- Medium** — Human escalation flow — specialist advice with IP attorney / AYUSH facilitation center contacts
- Medium** — Chat history + session memory (multi-turn conversation retention)
- Medium** — Bhashini multilingual pipeline — wire Hindi + Marathi language codes to Sarvam STT/TTS

Stage 3 — Production Hardening

~6–10 weeks after Stage 2. DPDPA, evaluation framework, deployment.

- 1
- 2
- 3
- 4
- 5
- 6
- 7

3 · Quick Wins (Start Immediately)

High-impact, low-effort tasks that can be completed in a single session:

- ✓ **Add jurisdiction toggle** — Single dropdown in chat-widget.js
- ✓ **Add legal disclaimer** — Modify `clean_answer()` or API response in `app.py`
- ✓ **Upgrade `data/ip_statutes/ip_statutes.txt`** — Add 2024 Patents Amendment & 2024 Biodiversity Rules text
- ✓ **Add WIPO GRATK Treaty (2024) content** — New file under `data/` — treaty provisions
- ✓ **Verify `chat-widget.css` / `chat-widget.js`** — Files referenced in `index.html` but may be missing
- ✓ **Surface confidence badge** — Already exists in API; just render in UI
- ✓ **Add `data/corpus_version.json`** — Version, date, source list for traceability

4 · Corpus Gaps — Topics With Zero Coverage

These topics appear in the specification but have no coverage in the current data folder:

Critical: 2024 Rules and WIPO GRATK Treaty are named in text but lack actual provisions.

Topic	Spec Section	Effort
WIPO Treaty on Genetic Resources and Traditional Knowledge (2024)	International	Medium
2024 Patents Amendment Act — full provisions	India IP	Medium
2024 Biodiversity Rules — full provisions	India ABS	Medium
PCT (Patent Cooperation Treaty) procedure	International	Medium
Madrid System for trademarks	International	Low-Medium
Hague System for designs	International	Low-Medium
Budapest Treaty (micro-organism deposits)	International	Medium
FSSAI Ayurveda-Aahar regulations (2022)	India Drug	Medium
Drugs and Magic Remedies Act (DMRDA) — detailed	India Drug	Low
DPCO (Drug Price Control Order) — detailed	India Drug	Low
EU herbal market access (THR / Directive 2002/46/EC)	International	High

5 · Verdict & Next Steps

What you have: A working MVP that demonstrates the core concept. The hardest parts — the RAG pipeline, voice input, multilingual TTS, and source attribution — are all in place. It is a solid **proof-of-concept that a citation-grounded Ayurveda IPR assistant is feasible.**

The remaining work is **content expansion, UI depth, and orchestration** — exactly what Stage 1 of the roadmap targets. **Stage 1 is achievable in 6–8 weeks** of focused development.

Recommended Next Actions (Priority Order)

1. Fix missing frontend files — create chat-widget.css and chat-widget.js
2. Add jurisdiction toggle + split answer display in the chat widget UI
3. Add legal disclaimer to every API response
4. Populate 2024 Patents Amendment Act and 2024 Biodiversity Rules text
5. Build the formulation classifier wizard — the most differentiating feature
6. Build the ABS compliance checklist wizard
7. Add TKDL pointer with actual database link
8. Add WIPO GRATK Treaty (2024) content
9. Wire up Hindi + Marathi language codes in app.py LANG_CODES map
10. Build evaluation test set from Questions.txt